

Q1 - 25 July - Shift 1

Let the locus of the centre (α, β) , $\beta > 0$, of the circle which touches the circle $x^2 + (y - 1)^2 = 1$ externally and also touches the x-axis be L. Then the area bounded by L and the line $y = 4$ is :

- (A) $\frac{32\sqrt{2}}{3}$ (B) $\frac{40\sqrt{2}}{3}$ (C) $\frac{64}{3}$ (D) $\frac{32}{3}$

Space for your notes:

Q2 - 25 July - Shift 2

If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3 - \sqrt{3})x + x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$, touch internally at the point $P(\alpha, \beta)$, then $(\alpha + \sqrt{3})^2 + (\beta + \sqrt{6})^2$ is equal to _____

Space for your notes:

Q3 - 27 July - Shift 1

If the circle $x^2 + y^2 - 2gx + 6y - 19c = 0$, $g, c \in \mathbb{R}$ passes through the point $(6, 1)$ and its centre lies on the line $x - 2cy = 8$, then the length of intercept made by the circle on x-axis is

- (A) $\sqrt{11}$ (B) 4
(C) 3 (D) $2\sqrt{23}$

Space for your notes:

Q4 - 27 July - Shift 2

Questions

MathonGo

A circle C_1 passes through the origin O and has diameter 4 on the positive x -axis. The line $y = 2x$ gives a chord OA of a circle C_1 . Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x -axis at P and y -axis at Q , then $QA : AP$ is equal to :

- (A) 1 : 4 (B) 1 : 5
(C) 2 : 5 (D) 1 : 3

*Space for your notes:***Q5 - 28 July - Shift 1**

Let C be the centre of the circle $x^2 + y^2 - x + 2y = \frac{11}{4}$ and P be a point on the circle. A line passes

through the point C , makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at the points Q and R . Then the area of the triangle PQR (in unit²) is :

- (A) 2 (B) $2\sqrt{2}$
(C) $8\sin\left(\frac{\pi}{8}\right)$ (D) $8\cos\left(\frac{\pi}{8}\right)$

*Space for your notes:***Q6 - 28 July - Shift 2**

Let the tangents at two points A and B on the circle $x^2 + y^2 - 4x + 3 = 0$ meet at origin $O(0, 0)$. Then the area of the triangle of OAB is

- (A) $\frac{3\sqrt{3}}{2}$ (B) $\frac{3\sqrt{3}}{4}$
(C) $\frac{3}{2\sqrt{3}}$ (D) $\frac{3}{4\sqrt{3}}$

*Space for your notes:***Q7 - 29 July - Shift 1**

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Questions

MathonGo

Let a line L pass through the point of intersection of the lines $bx + 10y - 8 = 0$ and $2x - 3y = 0$, $b \in \mathbb{R} - \left\{\frac{4}{3}\right\}$. If the line L also passes through the point $(1, 1)$ and touches the circle $17(x^2 + y^2) = 16$, then the eccentricity of the ellipse $\frac{x^2}{5} + \frac{y^2}{b^2} = 1$ is :

(A) $\frac{2}{\sqrt{5}}$

(B) $\sqrt{\frac{3}{5}}$

(C) $\frac{1}{\sqrt{5}}$

(D) $\sqrt{\frac{2}{5}}$

Space for your notes:

Q8 - 29 July - Shift 1

Let the mirror image of a circle $c_1 : x^2 + y^2 - 2x - 6y + \alpha = 0$ in line $y = x + 1$ be $c_2 : 5x^2 + 5y^2 + 10gx + 10fy + 38 = 0$. If r is the radius of circle c_2 , then $\alpha + 6r^2$ is equal to _____

Space for your notes:

Q9 - 29 July - Shift 2

Let AB be a chord of length 12 of the circle $(x - 2)^2 + (y + 1)^2 = \frac{169}{4}$.

Space for your notes:

If tangents drawn to the circle at points A and B intersect at the point P , then five times the distance of point P from chord AB is equal to ____.

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Answer Key

Q1 (C)

Q2 (25)

Q3 (D)

Q4 (A)

Q5 (B)

Q6 (B)

Q7 (B)

Q8 (12)

Q9 (72)

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Q1 (C)

$$(\alpha - 0)^2 + (\beta - 1)^2 = (\beta + 1)^2$$

$$\alpha^2 = 4\beta$$

$$x^2 = 4y$$

$$A = 2 \int_0^4 \left(4 - \frac{x^2}{4} \right) dx = \frac{64}{3}$$

Q2 (25)

(25)

Q3 (D)

Given circle $x^2 + y^2 - 2gx + 6y - 19c = 0$

Passes through (6, 1)

$$12g + 19c = 43 \quad \dots(1)$$

Centre (g, -3) lies on given line

$$\text{So, } g + 6c = 8 \quad \dots(2)$$

Solve equation (1) & (2)

$$c = 1 \text{ \& } g = 2$$

$$\text{equation of circle } x^2 + y^2 - 4x + 6y - 19 = 0$$

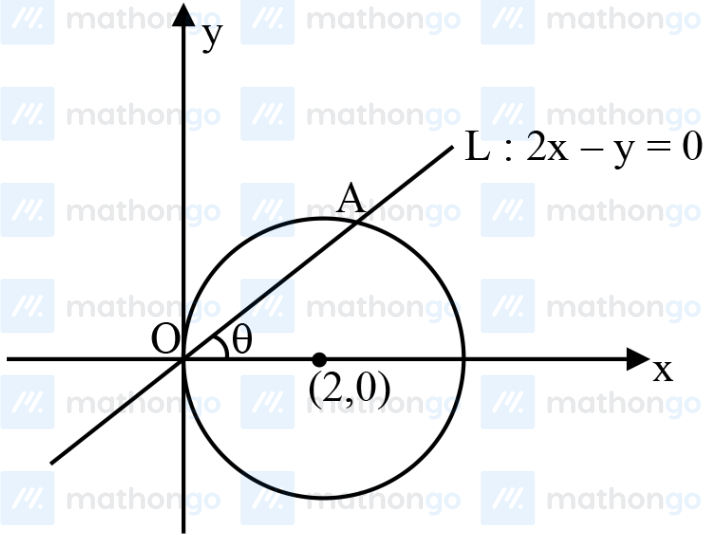
Length of intercept on x-axis

$$= 2\sqrt{g^2 - c} = 2\sqrt{23}$$

Q4 (A)

$$C_1 : x^2 + y^2 - 4x = 0$$

$$\tan \theta = 2$$

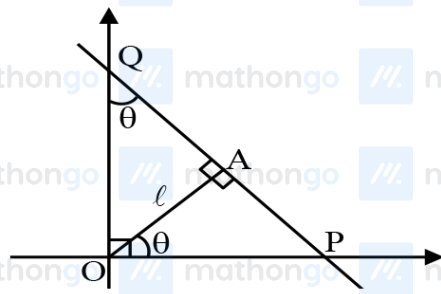


C_2 is a circle with OA as diameter.

So, tangent at A on C_2 is perpendicular to OR

Let $OA = \ell$

$$\begin{aligned} \therefore \frac{QA}{AP} &= \frac{\ell \cot \theta}{\ell \tan \theta} \\ &= \frac{1}{\tan^2 \theta} = \frac{1}{4} \end{aligned}$$



Q5 (B)

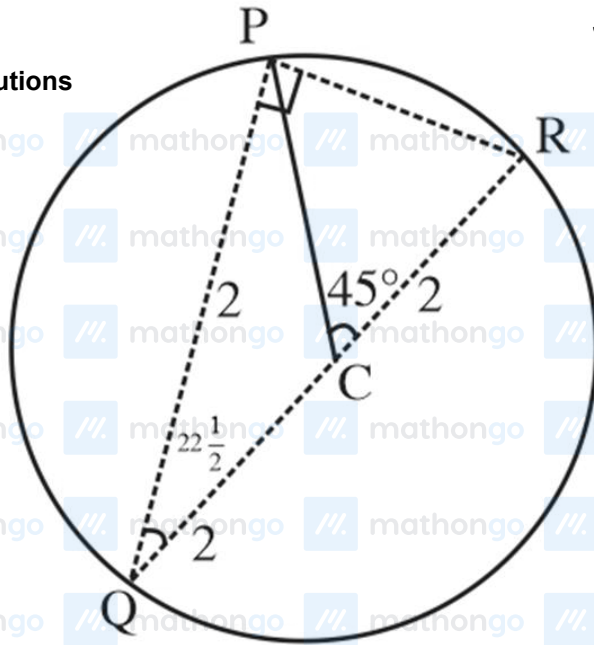
$$x^2 + y^2 - x + 2y = \frac{11}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + (y + 1)^2 = (2)^2$$

Or ΔPQR

$$PR = QK \sin 2 \geq \frac{1}{3}$$

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$$= 4 \cdot 6 \sin \frac{\pi}{8}$$

$$PQ = QR \cos 22\frac{1}{2}$$

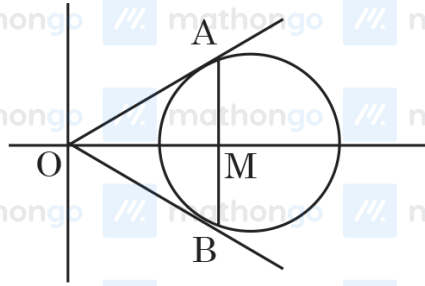
$$= 4 \cos \frac{\pi}{8}$$

$$\text{As } \Delta PQR = \frac{1}{2} PR \times PQ$$

$$= \frac{1}{2} \left(4^2 \sin \frac{\pi}{6} \right) \left(4 \cos \frac{\pi}{8} \right)$$

$$= 4 \sin \frac{\pi}{4} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

Q6 (B)



$$C : (x - 2)^2 + y^2 = 1$$

$$\text{Equation of chord AB : } 2x = 3$$

$$OA = OB = \sqrt{3}$$

$$AM = \frac{\sqrt{3}}{2}$$

$$\text{Area of triangle OAB} = \frac{1}{2}(2AM)(OM)$$

$$= \frac{3\sqrt{3}}{4} \text{ sq. units}$$

Q7 (B)

Line is passing through intersection of $bx + 10y - 8 = 0$ and $2x - 3y = 0$ is $(bx + 10y - 8) + \lambda(2x - 3y) = 0$. As line is passing through $(1, 1)$ so $\lambda = b + 2$

Now line $(3b + 4)x - (3b - 4)y - 8 = 0$ is tangent to circle $17(x^2 + y^2) = 16$

$$\text{So } \frac{8}{\sqrt{(3b+4)^2 + (3b-4)^2}} = \frac{4}{\sqrt{17}}$$

$$\Rightarrow b^2 = 2 \Rightarrow e = \sqrt{\frac{3}{5}}$$

Q8 (12)

Image of centre $c_1 \equiv (1, 3)$ in $x - y + 1 = 0$ is given

by

$$\frac{x_1 - 1}{1} = \frac{y_1 - 3}{-1} = \frac{-2(1 - 3 + 1)}{1^2 + 1^2}$$

$$\Rightarrow x_1 = 2, y_1 = 2$$

$\therefore$ Centre of circle $c_2 \equiv (2, 2)$

$$\therefore \text{Equation of } c_2 \text{ be } x^2 + y^2 - 4x - 4y + \frac{38}{5} = 0$$

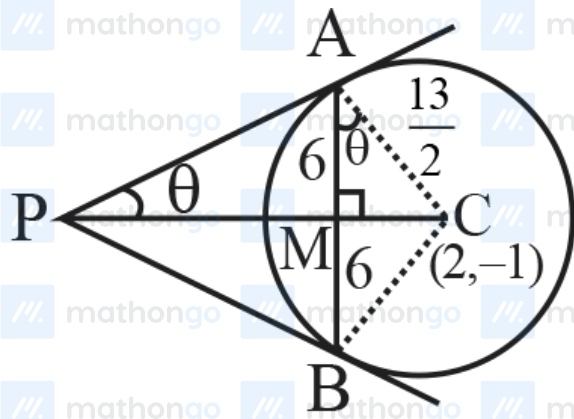
$$\text{Now radius of } c_2 \text{ is } \sqrt{4 + 4 - \frac{38}{5}} = \sqrt{\frac{2}{5}} = r$$

$$(\text{radius of } c_1)^2 = (\text{radius of } c_2)^2$$

$$\Rightarrow 10 - \alpha = \frac{2}{5} \Rightarrow \alpha = \frac{48}{5}$$

$$\therefore \alpha + 6r^2 = \frac{48}{5} + \frac{12}{5} = 12$$

Q9 (72)



$$\cos \theta = \frac{6}{13} = \frac{12}{26}$$

$$\sin \theta = \frac{5}{13}$$

$$PM = AM \cot \theta$$

$$PM = 6 \left(\frac{12}{5} \right) \therefore 5(PM) = 72$$